Capstone Project: Make the Computer Draw a Box

Nand to Tetris – I (ETCCNT301) — School of Engineering & Technology, K.R. Mangalam University

Marks: 10 — Individual work — COs: CO1–CO4 — Deadline: _____

1. The idea in one line

You will make the Hack computer **show a black box on the screen while a key is pressed**. You will do it in three small parts, using what you learned in the course.

Personal number K = last two digits of your roll number (e.g. 2501420041 gives K = 41). The box position and one assembler feature depend on K, so every student's work is a little different. Write your K on the first page of your report.

2. Part A – Build the computer

- Join your lab chips (gates, ALU, registers, RAM, PC) into `CPU.hdl`, `Memory.hdl` and `Computer.hdl`.
- Check them with the official test files (`CPU.tst`, `Memory.tst`, `ComputerAdd.tst`, `ComputerMax.tst`, `ComputerRect.tst`) and take a screenshot of each result.
- If one of your older chips does not work, you may use the built-in chip, but mention it in your report.

3. Part B – Write a small assembler in Python

The assembler turns a Hack program (`.asm`) into 0s and 1s (`.hack`).

- **First reading:** remove comments and blank lines, and note where each label like `(LOOP)` is.
- **Second reading:** turn every line into 16 bits. New variables get RAM addresses 16, 17, 18 and so on.
- Test on `Add.asm`, `Max.asm`, `Rect.asm` and `Pong.asm`. The output must match the official assembler.
- **One extra feature, based on K:**

K mod 3	Extra feature
0	Save all labels and variables with their addresses in `symbols.txt`.
1	For a wrong line, print a clear message with the line number (e.g. `Line 7: invalid instruction`) instead of crashing.
2	Print a warning if the same label is written twice.

4. Part C – Write the box program

Write `Box.asm` in Hack assembly. While any key is pressed, a black box appears. When the key is released, the box disappears.

- **Your box:** starts at row K, is 20 rows tall, and is $16 \times ((K \bmod 4) + 1)$ pixels wide from the left edge.
- Convert it with your **own** assembler from Part B (not the official one), run it in the CPU Emulator, and take a screenshot of your box.

Hint: One screen row = 32 words, and one word = 16 pixels. Row r starts at address `SCREEN + 32*r`. This is the Coursera `Fill.asm` program, drawn on a smaller area. Build it in small steps: one dash → wider dash → box with a loop → add the key check.

5. What to submit

1. **GitHub:** a `capstone/` folder in your existing lab repository with the HDL files, `assembler.py`, `Box.asm` and `Box.hack`.
2. **Short report (3–4 pages) on ZOSIMA:** name, roll no., K, box size, test screenshots, a screenshot of the box, and one paragraph about one mistake you fixed.

6. Viva (5 minutes)

You will open your own files, explain them, and make one small change on the spot (for example, moving the box down). If you cannot explain a part, marks for that part may not be given.

7. Marks (10)

Part	Marks
A. Computer passes the official tests	2.5
B. Assembler gives correct output (2.5) + extra feature for your K (0.5)	3
C. Correct box for your K, made with your own assembler	2.5
Viva	2
Total	**10**